LLMs as Resonant Structures

From Green's Functions to Nonlinear Spectral Theory

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Abstract

We develop a spectral theory of large language models in three layers. First, the classical picture: an LLM is a resonant structure whose Green's function encodes its natural modes—a prompt excites the structure, and the response is determined by the interaction between forcing function and eigenmodes. Second, the nonlinear correction: the Inverse Scattering Transform shows that certain nonlinear systems are "secretly linear" via a nonlinear change of basis, and sparse autoencoders (SAEs) perform exactly this role for LLM activations. Third, the global obstruction: the local linearisations on competence patches form a presheaf on the latent manifold, and polysemanticity is the failure of the gluing axiom—measurable by sheaf cohomology. Each level subsumes the last. The boundary between "linear" and "nonlinear" is not a wall. It is a choice of basis.

1. The Punchline

"Linear" versus "nonlinear" is not a property of the physics. It is a property of your coordinate system.

The Korteweg–de Vries equation, governing shallow water waves, looks nonlinear in physical coordinates. In scattering-data coordinates, it is as linear as the heat equation. The nonlinearity was never in the waves. It was in the description.

The same is true of large language models. An LLM's internal dynamics are massively nonlinear—attention, layer normalisation, GeLU activations. But sparse autoencoders reveal that the activations decompose into interpretable, approximately independent features. In the feature basis, the representation is approximately diagonal. The nonlinearity is absorbed into the change of coordinates.

This paper develops the analogy in three stages, each subsuming the last:

1. **Green's functions.** A linear spectral decomposition of the LLM as a resonant structure with natural modes. Valid locally, and a good first approximation.
2. **The Inverse Scattering Transform.** A nonlinear spectral theory that explains how SAEs find hidden linearity in a nonlinear system—the same trick that makes solitons "secretly linear."
3. **Sheaf structure.** The local linearisations form a presheaf on the latent manifold. Polysemanticity is the failure of the gluing axiom. The first cohomology H^1 measures the degree of context-dependence.

2. Green's Functions: The Linear Theory

2.1. Impulse responses

Given a linear differential operator L on a domain Ω, the Green's function $G(x, y)$ solves

$$L\,G(\cdot, y) = \delta(y),$$

where $\delta(y)$ is the Dirac delta centred at y. This is the system's response to a point impulse: strike the drum at exactly one point and observe how the entire membrane responds.

The power of the Green's function lies in *superposition*. Any input f decomposes as $f(x) = \int_\Omega f(y)\,\delta(y)\,dy$, so the solution to $Lu = f$ is

$$u(x) = \int_\Omega G(x, y)\, f(y)\, dy.$$

The operator L is the fixed structure—the drum, the membrane, the resonant cavity. The Green's function encodes everything about what that structure *can do*. The input f merely selects which modes get excited.

2.2. Spectral decomposition

If L has eigenfunctions φ_n with eigenvalues λ_n, then:

$$G(x, y) = \sum_n \frac{\varphi_n(x)\,\varphi_n(y)}{\lambda_n}.$$

Each eigenmode φ_n is a natural vibration pattern of the system. A small eigenvalue λ_n means the corresponding mode resonates strongly—it takes little energy to excite it. The input $f = \sum c_n \varphi_n$ simply sets the amplitudes: the response is $u = \sum (c_n/\lambda_n)\,\varphi_n$.

2.3. The LLM as resonant structure

Calling a large language model a "next-token predictor" is technically correct and completely beside the point. The training objective (next-token prediction) is the *bow*—the mechanism that shaped the instrument during training. What emerged is the instrument itself: a resonant structure with natural modes, preferred patterns of response, directions in activation space that ring when struck.

A prompt does not "instruct" the LLM. It *excites* it—a forcing function applied to a pre-existing vibrational structure. The response is determined by the interaction between the forcing function (prompt) and the natural modes (learned representations).

PDE / Green's function	LLM
Linear operator L	Trained network (weights fixed at inference)
Domain Ω	Token/embedding space
Green's function $G(x, y)$	Response map: prompt $\to$ output distribution
Eigenmodes φ_n	SAE features / representation directions
Eigenvalues λ_n	Feature activation thresholds
Input $f(x)$	Prompt
Solution $u = G * f$	Completion
Resonance	Prompt strongly activating a coherent feature

This picture already illuminates several phenomena. Prompt engineering becomes spectral engineering: constructing a forcing function f that excites the right modes. Fine-tuning changes L (reshapes the cavity); prompting changes f (excites it differently). RLHF reshapes the cavity to amplify some modes and damp others—the base model and the chat model are different instruments.

This is the linear story. It is correct as far as it goes. But LLMs are not linear.

3. The Nonlinearity Objection, Taken Seriously

Green's functions decompose the response into harmonic modes. But purely harmonic decompositions have well-known limitations: they struggle with shock waves, jets, and sharp impulses. They approximate these phenomena, but there are more efficient descriptions. In the language of signal processing, Fourier analysis assumes global smoothness that nonlinear systems violate.

The standard response—"the network is locally linear around any operating point, so we can compute a local Jacobian"—is true but insufficient. Local linearisation is the content of the Hartman–Grobman theorem, and it works near equilibria. But the interesting behaviour of LLMs happens far from equilibrium: when the model is composing concepts, resolving ambiguity, or navigating between incompatible interpretations.

PCA applied to LLM activations is the linear spectral theory—the Green's function decomposition. It captures the dominant modes but misses the nonlinear interactions between them. Sparse autoencoders outperform PCA precisely because they apply a *nonlinear* decomposition that can separate features packed into fewer dimensions than there are features.

What we need is not a defence of linearity. We need a nonlinear spectral theory.

4. Solitons: The Nonlinear Spectral Theory

4.1. The Inverse Scattering Transform

The Korteweg–de Vries (KdV) equation

$$u_t + 6uu_x + u_{xxx} = 0$$

governs shallow water waves. It is nonlinear (the uu_x term). Its soliton solutions—localised waves that propagate without changing shape and survive collisions intact—were among the great surprises of 20th-century mathematical physics.

The surprise deepened when Gardner, Greene, Kruskal, and Miura discovered the Inverse Scattering Transform (IST) in 1967. The IST solves KdV using exactly the same pipeline as Green's functions solve linear PDEs:

Linear PDE (heat equation)	**Nonlinear PDE (KdV)**
Fourier transform the initial data	Compute scattering data of $u(x, 0)$
Each mode evolves independently	Each piece evolves independently
(multiply by $e^{-k^2 t}$)	(multiply by simple exponentials)
Inverse Fourier transform	Inverse scattering

The scattering data consists of two parts: a *discrete spectrum* (finitely many eigenvalues $\lambda_1, \ldots, \lambda_N$, each with a norming constant c_n) and a *continuous spectrum* (a reflection coefficient $R(k)$). Each discrete eigenvalue corresponds to exactly one soliton. Its eigenvalue determines its speed and amplitude. A 3-soliton solution has exactly 3 discrete modes.

The key fact: the eigenvalues do not change under KdV flow. They are conserved quantities. Solitons survive collisions not because of some mysterious nonlinear balancing act, but because eigenvalues are invariants of the evolution. In scattering-data coordinates, KdV is as linear as the heat equation.

The nonlinearity was never in the physics. It was in the choice of coordinates.

4.2. SAEs as the Inverse Scattering Transform for LLMs

The analogy between the IST and sparse autoencoders is structural, not metaphorical:

IST (soliton theory)	**SAE (LLM interpretability)**
Wave profile $u(x)$	Activation vector in residual stream
Scattering transform	SAE encoder (nonlinear: ReLU)
Discrete eigenvalues (solitons)	Monosemantic features
Continuous spectrum (radiation)	Polysemantic residual
Inverse scattering	SAE decoder
Eigenvalues are invariants	Features persist across layers
Nonlinear PDE (KdV)	Nonlinear transformer computation

In both cases, the essential move is the same: a nonlinear change of basis that reveals hidden linearity. The scattering transform takes a nonlinear wave and decomposes it into modes that evolve independently. The SAE encoder takes a superposition of entangled features and decomposes it into sparse, approximately independent activations.

This explains why SAEs succeed where PCA fails. PCA is the Fourier transform—a *linear* change of basis. It works when the modes are orthogonal and the dynamics are linear. But the superposition hypothesis tells us that LLMs pack more features than dimensions, so features are non-orthogonal. You need a *nonlinear* decomposition to untangle them, just as you need the IST (not the Fourier transform) to decompose a nonlinear wave.

PCA is the Green's function. SAEs are the Inverse Scattering Transform.

4.3. The three regimes

Not all nonlinear systems admit the IST. The ones that do are called *integrable*. The analogy extends to a classification:

Regime	PDE analogue	LLM analogue
Integrable	Solitons (clean discrete spectrum)	Fully monosemantic model
Nearly integrable	Perturbed solitons (slow radiation)	Mostly monosemantic, some polysemantic
Non-integrable	Turbulence (no clean decomposition)	Heavily polysemantic

Real LLMs sit in the middle row. SAEs work well enough to extract many clean features, but the decomposition is never perfect. The residual polysemanticity is the continuous spectrum—the radiation that the IST separates from the solitons but cannot eliminate.

5. The Sheaf Structure

The IST provides a global linearisation when one exists. But what happens when the system is only *locally* linearisable—when different regions of input space admit different clean decompositions that may be mutually incompatible?

This is the situation for real LLMs, and the right language for it is sheaf theory.

5.1. The presheaf of local linearisations

Let M denote the LLM's latent manifold (the space of activation vectors, or equivalently, the space of inputs under the embedding map). Give M its natural topology: nearby points are inputs that produce similar activations.

Define a presheaf $\mathcal{F}$ on M as follows:

- To each open set $U \subseteq M$, assign $\mathcal{F}(U) =$ the set of local linearisations on U. Concretely, this is a local SAE decomposition: a set of feature directions that cleanly decompose all activations in U into sparse, interpretable components.
- For $V \subseteq U$, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ restricts the decomposition to the smaller region. If a linearisation works on U, it works on any $V \subseteq U$.

This is always at least a presheaf. The question is whether it satisfies the sheaf axioms.

5.2. Polysemanticity is the failure of gluing

A sheaf requires two conditions beyond being a presheaf:

Locality (separation). If two local linearisations agree on every piece of an open cover, they are the same linearisation. This holds: if two SAE decompositions produce identical features on every sub-region, they are the same decomposition.

Gluing. If local linearisations on overlapping patches $\{U_i\}$ agree on all intersections $U_i \cap U_j$, they glue to a global linearisation on $\bigcup U_i$. This is where it fails.

A feature that activates as "financial institution" on patch U_1 (finance context) and "river edge" on patch U_2 (geography context) provides two local sections that cannot be glued into a coherent global section. The word "bank" has a clean, monosemantic meaning locally on each patch. But the local meanings are incompatible—they do not extend to a global feature.

This is polysemanticity, stated sheaf-theoretically: the presheaf $\mathcal{F}$ fails the gluing axiom. Local linearisations exist but are globally incompatible.

5.3. Cohomology measures polysemanticity

The obstruction to gluing lives in the first Čech cohomology $H^1(M, \mathcal{F})$. This is a standard construction: given an open cover $\{U_i\}$ and compatible local sections on each U_i, H^1 measures whether they extend to a global section.

- $H^1 = 0$: every compatible family of local linearisations glues to a global one. The presheaf is a sheaf. The model is globally monosemantic.
- $H^1 \neq 0$: there exist compatible local sections that do not glue. The model has irreducible polysemanticity—context-dependence that cannot be resolved by choosing better features.

The dimension of H^1 measures *how polysemantic* the model is: how many independent ways the local linearisations can fail to be globally coherent.

5.4. Holonomy and the Laplace–Beltrami connection

If we equip M with a Riemannian metric (induced, for instance, by the Fisher information of the output distribution), we can study its differential geometry. The Laplace–Beltrami operator Δ_g on M generalises the Laplacian; its eigenfunctions are the natural vibration modes of the manifold.

On a flat manifold, these reduce to Fourier modes—the Green's function picture. On a curved manifold, they capture the geometry of the representation space. Competence at a task corresponds to the manifold being locally flat in the relevant region: the eigenfunctions locally look like Fourier modes, and the dynamics are locally linear.

The holonomy of the manifold—what happens when you parallel-transport a feature vector around a closed loop in input space—measures the global curvature. If the holonomy is trivial (the vector returns unchanged), the local frames are globally compatible: the presheaf is a sheaf. If the holonomy is non-trivial, the vector returns rotated: the same feature means something different after traversing a loop through different contexts. This is polysemanticity as parallel transport failure.

A training objective that minimises the holonomy of the Laplace–Beltrami connection is, in sheaf-theoretic terms, attempting to *sheafify* the representation: to make local linearisations globally compatible by flattening the manifold via gradient descent.

5.5. Should we kill H^1?

There is a tension here worth preserving. The superposition hypothesis says that LLMs deliberately pack more features than dimensions—features are non-orthogonal precisely because this is efficient compression. A perfectly sheafified model (one where every local linearisation glues

globally) would need representations as high-dimensional as the feature space. The polysemanticity is not a bug; it is a compression strategy.

The right question may not be "make $H^1 = 0$" but rather "understand H^1 well enough to navigate it." Just as an engineer does not need to eliminate all nonlinearity from a system—only to understand which nonlinearities matter and when—an interpretability researcher does not need to eliminate polysemanticity, only to characterise when and how features change meaning across contexts.

6. Case Study: The RSK Transformer

We now present a concrete case where the spectral theory is directly observable: a small transformer trained to invert combinatorial bijections [1]. In the language developed above, this is an *integrable* system—one where the presheaf is a sheaf, the decomposition is global, and the full spectral structure can be recovered.

6.1. Setup

The Robinson–Schensted–Knuth (RSK) correspondence is a fundamental bijection in algebraic combinatorics mapping permutations $\sigma \in S_n$ to pairs (P, Q) of standard Young tableaux of the same shape. The *RSK transformer* is a 1.2M-parameter encoder that learns the inverse: given (P, Q), recover σ.

The architecture is a 6-layer transformer encoder ($d = 128$, 8 heads) with structured 2D token embeddings:

$$\mathrm{emb}(v, r, c, t) = E_{\mathrm{val}}(v) + E_{\mathrm{row}}(r) + E_{\mathrm{col}}(c) + E_{\mathrm{tab}}(t),$$

where v is the entry value, (r, c) its position in the tableau, and $t \in \{P, Q\}$ identifies which tableau it belongs to. The same architecture, without modification, also learns the Hillman–Grassl and Burge correspondences.

6.2. The drum shape: 2D embeddings as domain geometry

The ablation study reveals a spectral geometry result. The 2D structured embeddings define the geometry of the domain Ω—the shape of the drum:

Embedding structure	**Accuracy**	**Spectral interpretation**
Full 2D (row + col + tableau)	100%	Correct drum shape ⇒ correct eigenmodes
1D sequential position	72%	Wrong geometry ⇒ wrong modes, some survive
No tableau identity (P vs Q)	6%	Domain halved ⇒ lost half the spectrum
No spatial information	0%	No domain ⇒ no modes ⇒ no response

This is a direct analogue of Kac's question "Can one hear the shape of a drum?" [2]. Removing the 2D structure changes the drum from a two-dimensional membrane to a one-dimensional string: some modes survive (72%), but those encoding the 2D tableau geometry are destroyed. Removing the P-vs-Q identity glues two drums together, making the network unable to distinguish which half is vibrating.

6.3. SAE features as eigenmodes

Sparse autoencoder analysis reveals interpretable features corresponding precisely to the natural modes φ_n:

- **Insertion-order detectors**—temporal eigenmodes that activate when the network recognises which step of the RSK algorithm it is inverting.
- **Step-specific locators**—spatial eigenmodes that fire for specific tableau cell positions (r, c).
- **Burge rule features**—algebraic eigenmodes encoding the local recurrence $G(i,j) = \max(G(i-1,j),\ G(i,j-$ $G(i-1,j-1) + a_{ij}$.

These are not abstract directions requiring post hoc interpretation—they are recognisable steps of known algorithms, recovered as the natural basis of the network's internal representation.

6.4. Algorithmic generalisation as resonance

Training on only 0.00004% of S_{15} yields 99.99% exact-match accuracy on held-out data. In the Green's function picture, this is expected: you do not need to strike a drum at every point to determine its eigenmodes. A sparse set of impulses suffices to identify the spectral structure, because each eigenfunction is defined over the entire domain. Once the modes are learned, any input decomposes into the same basis.

The MLP baseline collapses from 95.67% at $n = 10$ to 3.07% at $n = 15$: an MLP has no spatial structure to support extended eigenmodes. It is a lookup table pretending to be a drum.

6.5. The RSK transformer as an integrable system

In the framework developed above, the RSK transformer sits in the *integrable* regime. The evidence:

- The SAE decomposition is global: the same features work across the entire input space (S_{15}). There is no polysemanticity—each feature has one meaning everywhere. The presheaf is a sheaf. $H^1 = 0$.
- 100% accuracy means the spectral decomposition is complete: no information is lost in the radiation/residual.
- The ablation study shows what happens when you *break* the sheaf structure: removing geometric information (the domain topology) introduces curvature into the representation, forces polysemanticity, and accuracy degrades.
- The MLP baseline is a non-integrable system: no spatial structure to support coherent modes, high effective H^1, poor generalisation.

The RSK transformer offers a laboratory for the spectral theory of neural networks precisely because it is small enough to be fully interpretable yet rich enough to learn deep algebraic structure—and integrable enough that the full spectral picture is accessible.

7. Conclusion: The Three Levels

The spectral theory of LLMs has three levels, each answering a question the previous cannot:

Level	Mathematics	LLM tool	What it explains
1. Linear	Green's functions	PCA	Natural modes exist
2. Nonlinear	Inverse Scattering	SAEs	Hidden linearity in nonlinear systems
3. Global	Sheaf cohomology	H^1 analysis	Why local decompositions fail globally

At Level 1, the LLM is a drum. Prompts excite its natural frequencies. This is true, useful, and incomplete.

At Level 2, the drum is nonlinear—but "nonlinear" is a property of the coordinates, not the physics. SAEs find the scattering-data basis in which the representation becomes diagonal. Solitons survive collisions because eigenvalues are invariants; monosemantic features persist across layers for the same reason.

At Level 3, we confront the global question. Local linearisations exist on competence patches, but they may not glue. Polysemanticity is the cohomological obstruction. The dimension of H^1 measures how context-dependent the model's representations truly are—and whether that context-dependence is a defect to be repaired or a compression strategy to be understood.

An LLM is not a machine that follows instructions. It is a medium with a physics—and that physics has a spectral theory.

References

[1] R. Langer. *RSK Transformer: Learning inverse combinatorial bijections*. https://huggingface.co/RobBobin/rsk-transformer, 2025.

[2] M. Kac. Can one hear the shape of a drum? *The American Mathematical Monthly*, 73(4):1–23, 1966.